

Project Title

User Behaviour & Conversion Intelligence: A Full-Stack A/B Experiment Analytics Project

(Professional, interview-friendly, not weird like just “A/B Testing”)

Problem Statement

A digital platform launched an experiment to improve user engagement and conversion rates.

The objective of this project is to:

- Analyze user behaviour across sessions
 - Measure conversion performance across experiment groups
 - Identify high-performing traffic sources and payment methods
 - Evaluate revenue impact of user segments
 - Statistically validate whether experimental changes improved performance
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Dataset Summary

- Total Records: **10,000 transactions**
- Users segmented into: **New, Returning, VIP**
- Experiment Groups: **A & B**
- Traffic Sources: Organic, Paid, Referral, Social
- Devices: Desktop, Mobile, Tablet
- Payment Methods: Credit, Debit, UPI, Wallet
- Metrics: Revenue, Conversion, Session Time, Items Viewed

Data Structure Modeled Into:

- users_dim
 - session_dim
 - transaction_fact
-

● TOOL 1: Excel (Data Structuring & Business Summary Layer)

What We Did in Excel:

- Created Pivot Tables for:
 - Revenue by Group
 - Revenue by Traffic Source

- Conversions by Customer Segment
- Transactions by Payment Method
- Built quick executive-level summary metrics
- Validated totals before moving to BI tools

Why Excel Was Used:

Excel was used for:

- Fast business validation
- Quick KPI validation
- Executive-friendly static summaries

What to Keep from Excel:

Total revenue total conversion: Number of transactions		
₹ 19,20,046	6978	10000

group	Sum of revenue
A	950426.01
B	969619.72
Grand Total	1920045.73

Customer segment	Count of transaction_id
New	3354
Returning	3347
VIP	3299
Grand Total	10000

TOOL 2: Python (EDA + Statistical Validation Layer)

What We Did in Python:

Exploratory Data Analysis (EDA):

- Revenue distribution analysis
- Conversion comparison by group
- Device-based performance analysis
- Traffic performance trends
- Revenue spread visualization

Used:

- Matplotlib
 - Seaborn
 - Pandas
 - Statistical functions
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Statistical Analysis:

- Defined Null Hypothesis (H0)
- Performed Independent T-Test
- Calculated p-value
- Compared Mean Revenue between Group A & B

Key Statistical Outcome:

- p-value > 0.05
- Failed to reject null hypothesis
- No statistically significant difference in revenue between A & B

Why Python Was Used:

Python enabled:

- Deeper statistical validation
- Data science level credibility
- Hypothesis testing beyond BI visuals

What to Keep from Python:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10000 entries, 0 to 9999

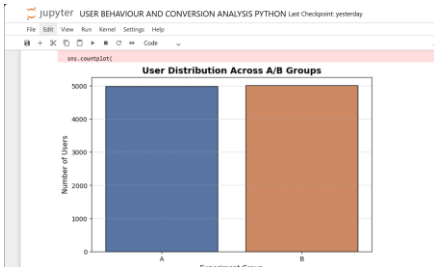
Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	session_id	10000 non-null	int64
1	user_id	10000 non-null	int64
2	group	10000 non-null	object
3	session_time_sec	10000 non-null	int64
4	num_items_viewed	10000 non-null	int64

- 5 device_type 10000 non-null object
- 6 region 10000 non-null object
- 7 transaction_id 10000 non-null int64
- 8 revenue 10000 non-null float64
- 9 conversion 10000 non-null int64
- 10 customer_segment 10000 non-null object
- 11 payment_method 10000 non-null object
- 12 traffic_source 10000 non-null object

dtypes: float64(1), int64(6), object(6)

memory usage: 1015.8+ KB



```
[16]: t_stat, p_value = ttest_ind(
      revenue_B,
      revenue_A,
      equal_var=False # Welch's t-test (safer for real-world data)
    )

      t_stat, p_value

[16]: (np.float64(0.8063635635937524), np.float64(0.42005242943151444))
```

Jupyter USER BEHAVIOUR AND CONVERSION ANALYSIS PYTHON

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Code

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STATISTICAL RESULTS & INTERPRETATION

A/B Test: Revenue Comparison

Summary of Test Results

Mean Revenue (Group A): 190.66

Mean Revenue (Group B): 193.34

t-statistic: 0.81

p-value: 0.42

Significance Level (α): 0.05

Hypothesis Testing Decision

Since: p-value (0.42) > α (0.05)

Fail to Reject the Null Hypothesis

📊 TOOL 3: SQL (Data Modeling + Business Intelligence Layer)

What We Did in SQL:

Database Design:

Created normalized schema:

- users_dim (Primary Key: user_id)
- session_dim (Primary Key: session_id, FK: user_id)
- transaction_fact (Primary Key: transaction_id, FK: session_id)

Schema Type:

👉 Snowflake-style dimension modeling

Business Queries Performed:

1. Basic Metrics Analysis
 - Total revenue
 - Conversion rate
 - ARPU
2. Funnel Analysis
 - Sessions → Transactions → Conversion
3. A/B Testing Performance Queries
 - Revenue by group
 - Conversion by device
 - Segment performance
4. Advanced Queries
 - CTEs
 - Window Functions
 - Aggregations
 - Ranking

Why SQL Was Used:

SQL demonstrated:

- Data modeling capability
- Relational thinking
- Analytical querying skills

- Production-level data handling

What to Keep from SQL:

The diagram shows a star schema with three tables: **transaction_fact**, **session_dim**, and **users_dim**.

- transaction_fact** (Fact Table):
 - transaction_id INT (Primary Key)
 - session_id INT (Foreign Key to session_dim)
 - revenue DECIMAL(10,2)
 - conversion TINYINT
 - payment_method VARCHAR(50)
- session_dim** (Dimension Table):
 - session_id INT (Primary Key)
 - user_id INT (Foreign Key to users_dim)
 - experiment_group CHAR(1)
 - device_type VARCHAR(50)
 - traffic_source VARCHAR(50)
 - session_time_sec INT
 - num_items_viewed INT
- users_dim** (Dimension Table):
 - user_id INT (Primary Key)
 - region VARCHAR(50)
 - customer_segment VARCHAR(50)

Relationships: **transaction_fact** is connected to **session_dim** (1:n). **session_dim** is connected to **users_dim** (1:n).

```

100 • SELECT
101     u.region,
102     s.experiment_group,
103     COUNT(t.transaction_id) AS total_orders,
104     ROUND(SUM(t.revenue),2) AS total_revenue
105 FROM users_dim u
106 JOIN session_dim s
107     ON u.user_id = s.user_id
108 JOIN transaction_fact t
109     ON s.session_id = t.session_id
110 GROUP BY u.region, s.experiment_group;
111
112
  
```

region	experiment_group	total_orders	total_revenue
East	A	1223	236583.29
South	B	1212	230349.26
South	A	1273	243569.40
West	A	1267	235850.93
North	B	1301	255983.71
West	B	1266	252428.77
North	A	1222	234422.39
East	B	1236	230857.98

Do NOT paste entire SQL file in README.

TOOL 4: Power BI (Executive Dashboard & Insight Layer)

What We Built:

Interactive Dashboard Including:

- KPI Cards (Revenue, Transactions, Conversion Rate)
- Revenue by Group
- Conversion by Device
- Traffic-wise Sessions

- Payment Method Revenue
- Segment Analysis
- Slicers for:
 - Group
 - Region
 - Traffic Source

Data Model:

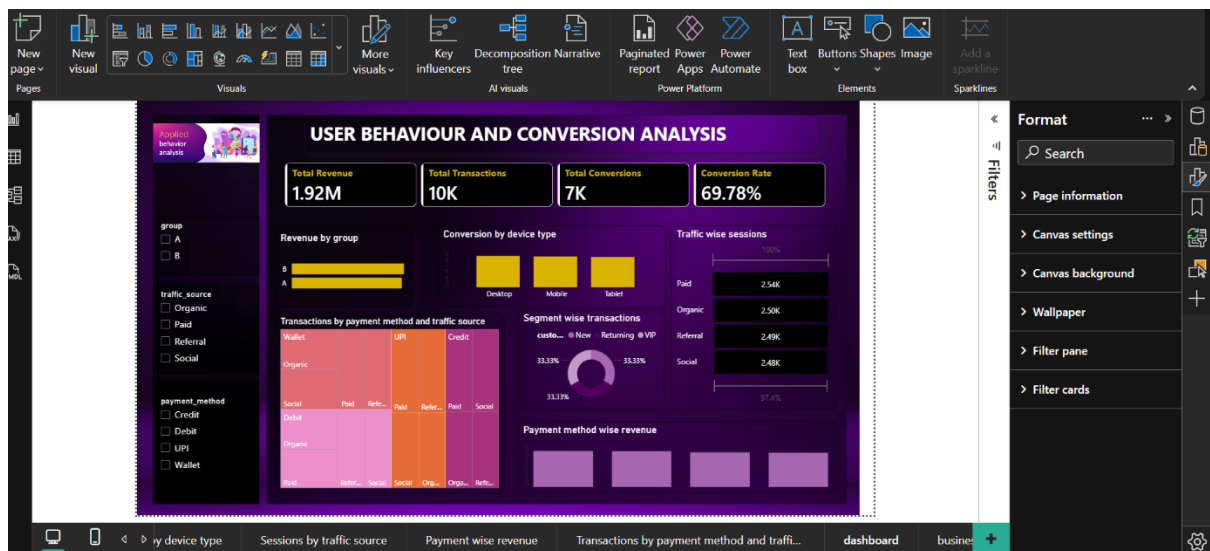
- One-to-Many relationships
- Proper star/snowflake design
- Fact connected to session, session connected to user

Why Power BI Was Used:

Power BI demonstrated:

- Data modeling skills
- Executive reporting ability
- Dashboard storytelling
- Business communication skills

What to Keep from Power BI:



Key Business Insights

- Group B generated slightly higher revenue but not statistically significant

- Paid traffic contributed highest revenue
 - VIP customers generated highest average revenue per transaction
 - Mobile users showed strongest conversion consistency
 - Wallet and Credit showed higher revenue stability
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Final Conclusion

Although Group B showed marginally higher revenue, statistical testing confirmed no significant experimental impact.

Business recommendation:

- Focus on high-value customer segments (VIP)
 - Optimize Paid traffic campaigns
 - Improve mobile experience
 - Enhance payment flow for top-performing payment methods
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Skills Demonstrated

- Data Cleaning & Validation (Excel)
- Statistical Testing (Python)
- Data Modeling & Querying (SQL)
- Dashboard Design & Storytelling (Power BI)
- Hypothesis Testing
- Funnel Analytics
- Business Insight Communication